Group 35
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DS4200 Project Description: This project analyzes Netflix's content strategy evolution from 2016 to 2021 using a combination of interactive data visualizations.

Project website link:
https://khanhhoang0712.github.io/ds4200-netflix-project/website/index.html

Design ideas:
1. Geographic Distribution of Netflix Content
Our first visualization is a choropleth world map that provides a geographic overview of Netflix's content distribution by country of origin. We place it at the beginning of the project so the audience can immediately understand the global scope of Netflix's catalog before diving into more specific trends.

The primary mark is area (geographic polygons), and color is used as the main channel to encode the number of Netflix titles produced per country. A sequential red color scale is used, ranging from light pink to dark red, to represent increasing content volume, which also aligns with Netflix's brand identity. A dropdown interaction allows users to filter the map by content type (All Titles, Movies Only, or TV Shows Only), enabling more targeted comparisons across regions. Hovering over any country reveals a tooltip with the exact title count, making precise values accessible without cluttering the map.

2. Netflix Content Growth and Type Evolution (2016–2021)
Our second visualization is a side-by-side line chart that combines two related trends into a single view. We chose to place these two charts together because they tell a connected story which is how much content Netflix added each year and what type of content made up that growth. Presenting them side by side allows the audience to compare the two trends simultaneously without switching between separate charts.

The primary mark is a line with points, and the y-axis encodes either total title count or percentage of content depending on the panel. On the right chart, color is used to distinguish between Movies and TV Shows, using Netflix red and dark red to maintain a consistent brand theme. A legend click interaction allows users to highlight or fade a specific content type, making it easier to focus on one trend at a time. Tooltips on hover provide exact values for each data point, enabling precise reading without cluttering the chart.

3. Netflix Genre Rankings Over Time (2016–2021)
Our third visualization is a bump chart that tracks how the top 8 genres on Netflix shifted in popularity ranking each year from 2016 to 2021. We chose a bump chart over a traditional bar chart or stacked area chart because it more clearly shows relative changes in ranking over time. The viewer can immediately see which genres rose, fell, or remained stable without being distracted by raw count differences between genres.

The primary mark is a line with points, where the y-axis encodes rank position (1 being most popular) and the x-axis encodes year. Color is used to distinguish each genre, using a multi-color categorical scheme since all 8 genres need to be clearly distinguishable from one another. A legend click interaction allows users to highlight a specific genre and fade the others, making it easier to trace any single genre's trajectory across the full time period. Tooltips on hover provide the exact rank and title count for each data point, giving precise context beyond what the visual position alone conveys

4. Distribution of Movie Budgets on Netflix (2016–2021)
Our fourth visualization is a layered chart that combines a boxplot with an average trend line to show how movie production budgets were distributed each year from 2016 to 2021. We chose a boxplot over a simple bar chart because it reveals not just the average but also the spread, variance, and outliers within each year.

The primary mark is a boxplot where the box encodes the interquartile range, the whiskers show the full spread, and the white line inside each box marks the median. An overlaid yellow line tracks the average budget per year. We chose a contrasting color so it stands out clearly against the red boxplots. Netflix red is used for the boxplots to maintain the consistent color theme across the project. Tooltips on hover provide exact budget values, allowing viewers to read precise numbers for any data point across both the boxplot and the average line.

5. Netflix Content Rating Distribution by Year (2016–2021)
Our fifth visualization is a heatmap that shows how the share of each content rating group changed across years from 2016 to 2021. We chose a heatmap over a stacked bar chart or grouped bar chart because it allows the viewer to compare patterns across both years and rating groups simultaneously in a compact grid format. The ordered rows also make it immediately clear how audience targeting shifts from youngest to most mature, which would be harder to read in a bar chart with multiple overlapping colors. The rating groups are ordered from Kids at the top to Mature at the bottom to reflect a natural progression from youngest to oldest audiences. Text labels are overlaid inside each cell to display the exact percentage, removing the need to reference a separate axis.

Feedback from the peer review:
The main feedback we got from the peer review is that the visualizations felt too similar to each other in chart type, relying heavily on bar charts, and that the interactions were not deep or varied enough to support meaningful exploration of the data. Reviews also noted that genre was missing as a dimension across the charts. We also got multiple feedback from the professor about the ordering of our charts as well as some details specific to our content rating distribution chart.

The changes we have made include replacing the average budget bar chart with a layered boxplot and average trend line to show budget variance over time, converting the ratings bar chart into a heatmap with sequential color and ordered rating categories, converting the movies vs TV shows bar chart into a side-by-side line chart combined with the total growth chart, and adding a brand new genre bump chart that tracks how the top 8 genres shifted in ranking from 2016 to 2021.

The reason we did not change the world map is that it was already a strong and unique visualization that provided a clear geographic overview, and peer feedback on it was mostly positive.